

Build & Deploy Guide

This project includes a **C# backend** (OpenAPI-based), an **Angular frontend**, and uses **Ollama**, **Apache Tika**, and **PostgreSQL** for various services. The development and deployment environments are fully containerized using Docker.

Prerequisites

- [Docker](https://www.docker.com/) (<https://www.docker.com/>)
 - [Docker Compose](https://docs.docker.com/compose/) (<https://docs.docker.com/compose/>)
 - [GitHub CLI \(optional\)](https://cli.github.com/) (<https://cli.github.com/>) (GitHub Workflows)
-

Getting Started Locally

Project Setup and Deployment Guide

1. Frontend-Backend Communication

- Build OpenAPI client Docker container and run it to set up the OpenAPI SDK in the frontend directory.
- Run the following commands:

```
cd BitAndBeam
docker compose build openapi-client
docker compose up openapi-client
```

- This sets up the `api` directory in `BitAndBeam/frontend/src`. The created functions should be used to make API calls to the backend from the frontend code.

2. Project Build

- To build and run the entire project:

```
cd BitAndBeam
docker compose build
docker compose up
```

3. Access the Services

- Frontend: <http://localhost:8080> (<http://localhost:8080>)
- Backend API: <http://localhost:5001> (<http://localhost:5001>)
- Swagger API docs: <http://localhost:5001/swagger> (<http://localhost:5001/swagger>)

4. Production Deployment

- Production deployment is automated through a GitHub workflow using `docker-compose-prod.yml`.
- The accessible links for the deployed services are available in the `README`.

5. Ollama Configuration

- Ollama can be configured using the `Dockerfile` and `entrypoint.sh` script in the `ollama` directory.
- Use `docker-compose-prod-ollama.yml` or `docker-compose-prod-ollama-gpu.yml` for deployment.
- Changes are automatically deployed to the Ollama server via GitHub workflows.

This will:

- Build and run the C# backend
- Serve the Angular frontend
- Start Ollama Service
- Launch Apache Tika (document parsing)
- Set up a PostgreSQL database instance

Accessing Local Services

- **Frontend:** `http://localhost:8080`
 - **Backend API:** `http://localhost:5001`
 - **PostgreSQL:** `localhost:5432` (credentials in `.env` or `docker-compose.yml`)
 - **Tika Server:** `http://localhost:9998`
 - **Ollama:** `http://localhost:11434` or remote server link
-

GitHub Actions CI/CD Pipeline

This project includes a **GitHub Actions** pipeline for automated builds and deployments to a server.

What It Does

- Runs on every push to the `main` branch
- Builds Docker images for each service
- Pushes them to the GitHub Container Registry
- Remotely runs containers on the production server

Workflow File

See `.github/workflows/docker-ci.yml` for the CI/CD definition.